Survival Analysis — Homework Solutions

February 20, 2026

Problem 1: Logit-Transformed Confidence Interval for the Survival Function (15 pts)

Logistic regression motivates the logit transformation $g(u) = \log\left(\frac{u}{1-u}\right)$ to improve finite-sample CI performance for the survival function.

(a) Variance of $g(\hat{S}(t))$

The derivative of the logit transformation is:

$$g'(u) = \frac{1}{u(1-u)}$$

By the **delta method**:

$$\boxed{\mathrm{Var}\big(g(\hat{S}(t))\big) \;\approx\; \big[g'(S(t))\big]^2\, V(\hat{S}(t)) \;=\; \frac{V(\hat{S}(t))}{\big[\hat{S}(t)\,(1-\hat{S}(t))\big]^2}}$$

(b) 95% CI for $g(S(t))$

By the large-sample normal approximation:

$$\boxed{g(\hat{S}(t)) \;\pm\; 1.96\,\frac{\sqrt{V(\hat{S}(t))}}{\hat{S}(t)\,(1-\hat{S}(t))}}$$

That is:

$$\log\frac{\hat{S}(t)}{1-\hat{S}(t)} \;\pm\; 1.96\,\frac{\sqrt{V(\hat{S}(t))}}{\hat{S}(t)\,(1-\hat{S}(t))}$$

Denote the lower and upper limits as L and U.

(c) 95% CI for $S(t)$

Apply the inverse logit $g^{-1}(x) = \frac{e^x}{1+e^x}$ to both endpoints:

$$\boxed{\left(\frac{e^L}{1+e^L}\,,\;\frac{e^U}{1+e^U}\right)}$$

This back-transformation guarantees the confidence interval stays within $(0, 1)$, which is the key advantage of using the logit transformation.

Problem 2: Bone Marrow Transplant Data — AML Low-Risk Group (30 pts)

Dataset: `bmt` from R package {`KMsurv`}.
Subgroup: AML Low-Risk (group = 2), using `t1` (time to death) and `d1` (death indicator).
Sample size: $n = 54$ (23 deaths, 31 censored).

(a) Kaplan–Meier Survival Function with Greenwood's SE

The Kaplan–Meier estimator:

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

Greenwood's formula:

$$\widehat{\text{Var}}(\hat{S}(t)) = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{n_i(n_i - d_i)}$$

Table 1: Kaplan–Meier Estimates for AML Low-Risk Group

Time	n_i	d_i	$\hat{S}(t)$	SE($\hat{S}(t)$)
10	54	1	0.9815	0.0183
35	53	1	0.9630	0.0257
48	52	1	0.9444	0.0312
53	51	1	0.9259	0.0356
79	50	1	0.9074	0.0394
80	49	1	0.8889	0.0428
105	48	1	0.8704	0.0457
222	47	1	0.8519	0.0483
288	46	1	0.8333	0.0507
390	45	1	0.8148	0.0529
393	44	1	0.7963	0.0548
414	43	1	0.7778	0.0566
431	42	1	0.7593	0.0582
481	41	1	0.7407	0.0596
522	40	1	0.7222	0.0610
583	39	1	0.7037	0.0621
641	38	1	0.6852	0.0632
653	37	1	0.6667	0.0642
704	36	1	0.6481	0.0650
1063	29	1	0.6258	0.0665
1074	28	1	0.6034	0.0678
1156	27	1	0.5811	0.0688
2204	6	1	0.4842	0.1054

(b) Nelson–Aalen Cumulative Hazard with SE

The Nelson–Aalen estimator and its standard error:

$$\tilde{H}(t) = \sum_{t_i \le t} \frac{d_i}{n_i}, \qquad \widehat{\text{SE}}(\tilde{H}(t)) = \sqrt{\sum_{t_i \le t} \frac{d_i}{n_i^2}}$$

Table 2: Nelson–Aalen Estimates for AML Low-Risk Group

Time	n_i	d_i	$\tilde{H}(t)$	SE($\tilde{H}(t)$)
10	54	1	0.0185	0.0185
35	53	1	0.0374	0.0264
48	52	1	0.0566	0.0327
53	51	1	0.0762	0.0381
79	50	1	0.0962	0.0430
80	49	1	0.1166	0.0476
105	48	1	0.1375	0.0520
222	47	1	0.1587	0.0562
288	46	1	0.1805	0.0602
390	45	1	0.2027	0.0642
393	44	1	0.2254	0.0681
414	43	1	0.2487	0.0720
431	42	1	0.2725	0.0758
481	41	1	0.2969	0.0796
522	40	1	0.3219	0.0835
583	39	1	0.3475	0.0873
641	38	1	0.3738	0.0912
653	37	1	0.4009	0.0951
704	36	1	0.4286	0.0991
1063	29	1	0.4631	0.1049
1074	28	1	0.4988	0.1108
1156	27	1	0.5359	0.1169
2204	6	1	0.7025	0.2036

(c) Mean Survival Time and 95% Linear CI

The restricted mean survival time (up to $\tau = 2569$ days):

$$\hat{\mu} = \int_0^\tau \hat{S}(t)\,dt = \sum_i \hat{S}(t_{i-1})(t_i - t_{i-1})$$

with variance:

$$\widehat{\text{Var}}(\hat{\mu}) = \sum_{t_i:\,d_i>0} \frac{A_i^2\,d_i}{n_i(n_i-d_i)}, \quad \text{where } A_i = \int_{t_i}^\tau \hat{S}(u)\,du$$

$$\hat{\mu} = \mathbf{1644.65} \text{ days}, \qquad \text{SE}(\hat{\mu}) = 147.09 \text{ days}$$

$$95\% \text{ Linear CI:} \quad \hat{\mu} \pm 1.96 \times 147.09 = \boxed{(1356.35,\ 1932.94) \text{ days}}$$

(d) Median Survival Time and 95% Log-Log Transformed CI

The estimated median is the smallest t where $\hat{S}(t) \le 0.5$:

$$\hat{S}(1156) = 0.5811 > 0.5, \quad \hat{S}(2204) = 0.4842 < 0.5 \implies \boxed{\text{Median} = 2204 \text{ days}}$$

For the **log-log transformed CI**, let $\theta(t) = \log(-\log(\hat{S}(t)))$ with:

$$\text{SE}(\hat{\theta}(t)) = \frac{\text{SE}(\hat{S}(t))}{\hat{S}(t)\,|\log \hat{S}(t)|}$$

The 95% log-log CI for $S(t)$ is $\left[\hat{S}(t)^{\exp(1.96\cdot\mathrm{SE}(\hat{\theta}))},\ \hat{S}(t)^{\exp(-1.96\cdot\mathrm{SE}(\hat{\theta}))}\right]$.

Time	$\hat{S}(t)$	SE	Log-log CI Lower	Log-log CI Upper
1063	0.6258	0.0665	0.4815	0.7404
1074	0.6034	0.0678	0.4580	0.7213
1156	0.5811	0.0688	0.4349	0.7019
2204	0.4842	0.1054	0.2709	0.6685

The lower CI bound for $S(t)$ drops below 0.5 at $t = 1063$ ($0.4815 < 0.5$), while the upper CI bound never drops below 0.5 within follow-up. Therefore:

$$\boxed{\text{95\% Log-Log CI for Median: } (1063,\ \infty)}$$

(e) 95% Arcsine-Square-Root CI for $S(300)$

At $t = 300$: $\hat{S}(300) = 0.8333$, $\mathrm{SE}(\hat{S}(300)) = 0.0507$.

The arcsine-square-root transformation $g(u) = \arcsin(\sqrt{u})$ with $g'(u) = \frac{1}{2\sqrt{u(1-u)}}$ gives:

$$\mathrm{SE}(g(\hat{S}(300))) = \frac{0.0507}{2\sqrt{0.8333 \times 0.1667}} = 0.0680$$

CI for $g(S(300))$:

$$1.1503 \pm 1.96 \times 0.0680 = (1.0169,\ 1.2836)$$

Back-transform using $g^{-1}(x) = \sin^2(x)$:

$$\boxed{\text{95\% CI for } S(300):\ (0.7233,\ 0.9198)}$$

(f) 95% EP Confidence Bands over [100, 400] days

The Equal Precision (EP) band (Nair, 1984) uses:

$$\text{Linear:} \quad \hat{S}(t) \pm c_\alpha \times \mathrm{SE}(\hat{S}(t))$$

$$\text{Log-log:} \quad \left[\hat{S}(t)^{\exp(c_\alpha\cdot\mathrm{SE}(\hat{\theta}))},\ \ \hat{S}(t)^{\exp(-c_\alpha\cdot\mathrm{SE}(\hat{\theta}))}\right]$$

where c_α is obtained from the distribution of $\sup_{\hat{\sigma}_L^2 \le u \le \hat{\sigma}_U^2} |W(u)|/\sqrt{u}$ with $\hat{\sigma}_L^2 = 0.002315$, $\hat{\sigma}_U^2 = 0.004737$.

By simulation (200,000 replicates): $c_\alpha = 2.498$ for 95% confidence.

Table 3: Linear EP Confidence Bands

Time	$\hat{S}(t)$	SE	EP Lower	EP Upper
105	0.8704	0.0457	0.7562	0.9845
222	0.8519	0.0483	0.7311	0.9726
288	0.8333	0.0507	0.7067	0.9600
390	0.8148	0.0529	0.6828	0.9468
393	0.7963	0.0548	0.6594	0.9332

Table 4: Log-Log Transformed EP Confidence Bands

Time	$\hat{S}(t)$	SE	EP Lower	EP Upper
105	0.8704	0.0457	0.6997	0.9475
222	0.8519	0.0483	0.6783	0.9359
288	0.8333	0.0507	0.6573	0.9238
390	0.8148	0.0529	0.6365	0.9113
393	0.7963	0.0548	0.6160	0.8984

Problem 3: Left-Truncated Survival of Diabetics (Exercise 4.7)

30 diabetic subjects recruited at a clinic; data is **left truncated** since subjects must survive from birth to their entry age to be observed.

(a) Number at Risk $Y(t)$ as a Function of Age

Subject i is at risk at age t if $\text{entry}_i < t \leq \text{exit}_i$.

Table 5: Risk Set as a Function of Age

Age	58	59	60	61	62	63	64	65	66	67	68	69	70
$Y(t)$	0	2	3	4	6	8	9	10	8	9	12	11	10
Deaths	0	0	1	0	1	1	0	2	1	0	2	2	2
Censored	0	0	0	0	0	0	0	0	0	0	0	2	0
Entries	2	1	2	2	3	2	1	0	2	3	1	3	3

Age	71	72	73	74	75	76	77	78	79	80
$Y(t)$	11	10	10	9	7	7	5	4	3	1
Deaths	2	2	1	1	0	0	1	0	0	0
Censored	0	1	1	1	0	2	0	1	2	1
Entries	1	3	1	0	0	0	0	0	0	0

Note: $Y(t)$ is *not* monotonically decreasing because new subjects enter the risk set at their entry (truncation) age.

(b) Conditional Survival Function $\hat{S}(t \mid T \geq 60)$

Using the product-limit estimator for left-truncated data:

$$\hat{S}(t \mid T \geq 60) = \prod_{\substack{t_i \leq t \\ t_i \geq 60}} \left(1 - \frac{d_i}{Y(t_i)}\right)$$

Table 6: $\hat{S}(t \mid T \geq 60)$

Age t	$Y(t)$	$d(t)$	$1 - d/Y$	$\hat{S}(t \mid T \geq 60)$	Decimal
60	3	1	2/3	2/3	0.6667
62	6	1	5/6	5/9	0.5556
63	8	1	7/8	35/72	0.4861
65	10	2	4/5	7/18	0.3889
66	8	1	7/8	49/144	0.3403
68	12	2	5/6	245/864	0.2836
69	11	2	9/11	245/1056	0.2320
70	10	2	4/5	49/264	0.1856
71	11	2	9/11	147/968	0.1519
72	10	2	4/5	147/1210	0.1215
73	10	1	9/10	1323/12100	0.1093
74	9	1	8/9	294/3025	0.0972
77	5	1	4/5	1176/15125	0.0778

(c) Conditional Survival Function $\hat{S}(t \mid T \geq 70)$

Table 7: $\hat{S}(t \mid T \geq 70)$

Age t	$Y(t)$	$d(t)$	$1 - d/Y$	$\hat{S}(t \mid T \geq 70)$	Decimal
70	10	2	4/5	4/5	0.8000
71	11	2	9/11	36/55	0.6545
72	10	2	4/5	144/275	0.5236
73	10	1	9/10	648/1375	0.4713
74	9	1	8/9	576/1375	0.4189
77	5	1	4/5	2304/6875	0.3351

Observation: At matching ages, the 70-year-old has substantially higher conditional survival than the 60-year-old (e.g., at age 74: 0.42 vs. 0.10), reflecting survivorship bias—those reaching 70 are a selected healthier subgroup.

Problem 4: Competing Risks — AML Low-Risk Group (15 pts)

Setup: Using disease-free survival time (`t2`) with two competing risks:

- **Cause 1 — Relapse** (`d2 = 1`): 9 events
- **Cause 2 — Death in remission** (`d3 = 1, d2 = 0`): 16 events
- Censored (`d3 = 0`): 29 subjects

The cumulative incidence function (CIF) for cause j:

$$\hat{F}_j(t) = \sum_{t_i \le t} \hat{S}(t_i^-) \cdot \frac{d_{ji}}{n_i}$$

where $\hat{S}(t^-)$ is the overall KM survival just before time t.

(a) Cumulative Incidence at One Year (365 days)

Table 8: Competing Risks Table (event times up to 1 year)

Time	n	d_1	d_2	$\hat{S}(t^-)$	h_1	h_2	$\hat{F}_1(t)$	$\hat{F}_2(t)$
10	54	0	1	1.0000	0	0.0185	0.0000	0.0185
35	53	0	1	0.9815	0	0.0189	0.0000	0.0370
48	52	0	1	0.9630	0	0.0192	0.0000	0.0556
53	51	0	1	0.9444	0	0.0196	0.0000	0.0741
79	50	0	1	0.9259	0	0.0200	0.0000	0.0926
80	49	0	1	0.9074	0	0.0204	0.0000	0.1111
105	48	0	1	0.8889	0	0.0208	0.0000	0.1296
211	47	1	0	0.8704	0.0213	0	0.0185	0.1296
219	46	1	0	0.8519	0.0217	0	0.0370	0.1296
248	45	1	0	0.8333	0.0222	0	0.0556	0.1296
272	44	1	0	0.8148	0.0227	0	0.0741	0.1296
288	43	0	1	0.7963	0	0.0233	0.0741	0.1481

At one year (365 days), no further events occur between $t = 288$ and $t = 365$:

$$\boxed{\hat{F}_{\text{relapse}}(365) = 0.0741, \qquad \hat{F}_{\text{death in remission}}(365) = 0.1481}$$

Verification: $\hat{F}_1(365) + \hat{F}_2(365) + \hat{S}(365) = 0.0741 + 0.1481 + 0.7778 = 1.0000$ ✓

(b) Standard Errors of the CIF Estimators

Using the Aalen (1978) variance formula:

$$\widehat{\text{Var}}(\hat{F}_j(t)) = \sum_{t_i \le t} \left[\frac{\hat{S}(t_i^-)^2 \, d_{ji}(n_i - d_{ji})}{n_i^3} + \frac{[\hat{F}_j(t) - \hat{F}_j(t_i)]^2 \, d_i}{n_i(n_i - d_i)} - \frac{2[\hat{F}_j(t) - \hat{F}_j(t_i)] \, \hat{S}(t_i^-) \, d_{ji}}{n_i^2} \right]$$

$$\boxed{\text{SE}(\hat{F}_{\text{relapse}}(365)) = 0.0356, \qquad \text{SE}(\hat{F}_{\text{death in remission}}(365)) = 0.0483}$$

(c) Conditional Probability at One Year

The conditional probability of each cause given failure by time t:

$$\hat{P}(\text{cause } j \mid T \leq t) = \frac{\hat{F}_j(t)}{1 - \hat{S}(t)} = \frac{\hat{F}_j(t)}{\hat{F}_1(t) + \hat{F}_2(t)}$$

With $1 - \hat{S}(365) = 0.2222$:

$$\boxed{\hat{P}(\text{Relapse} \mid T \leq 365) = \frac{0.0741}{0.2222} = \frac{1}{3} \approx 0.3333}$$

$$\boxed{\hat{P}(\text{Death in Remission} \mid T \leq 365) = \frac{0.1481}{0.2222} = \frac{2}{3} \approx 0.6667}$$

Interpretation: Among AML low-risk patients who fail within the first year, approximately one-third relapse and two-thirds die in remission. Death in remission is the dominant early competing risk.

Problem 5: Needlestick Injury — Mixed Censoring (Exercise 5.2)

100 veterinarians surveyed about time (months after graduation) to first needlestick injury. The data involves three types of observations:

- **Exact** at time t: interval $[t, t]$ — 27 subjects
- **Left-censored** at time t: interval $(0, t)$ — 32 subjects
- **Right-censored** at time t: interval (t, ∞) — 41 subjects

This is a **mixed-censoring** problem requiring the **Turnbull NPMLE** (EM algorithm).

Turnbull NPMLE — Estimated Survival Function

The EM algorithm converged after 417 iterations:

Table 9: Turnbull NPMLE for Needlestick Injury

Time (months)	$\hat{P}(T = t)$	$\hat{F}(t) = P(T \leq t)$	$\hat{S}(t) = P(T > t)$
2	0.0948	0.0948	0.9052
4	0.0632	0.1580	0.8420
8	0.0316	0.1897	0.8103
10	0.0542	0.2438	0.7562
12	0.0887	0.3325	0.6675
15	0.1174	0.4499	0.5501
20	0.0518	0.5017	0.4983
24	0.0487	0.5504	0.4496
28	0.0324	0.5828	0.4172
34	0.0167	0.5994	0.4006
86	0.3306	0.9300	0.0700

Key results:

- Zero mass at times 41, 62, 69, 75, 79 (no exact events observed).
- Total probability sums to 0.93; $\hat{S}(\infty) \approx 0.07$ ($\sim$7% may never experience injury).
- Large mass at $t = 86$ (0.331) absorbs probability from many late left-censored observations.

Estimated quantiles:

$$Q_1 = 12 \text{ months}, \qquad \text{Median} = 20 \text{ months}, \qquad Q_3 = 86 \text{ months}$$

Problem 6: Cohort Life Table for HIV Incidence (Exercise 5.8)

100 high-risk STD-infected, HIV-negative individuals (1980) followed with interval-censored HIV detection. Actuarial (life table) method with 2-year intervals.

Notation:

- $n'_i = n_i - w_i/2$ = effective number at risk (actuarial adjustment)
- $\hat{q}_i = d_i/n'_i$ = conditional probability of HIV acquisition
- $\hat{p}_i = 1 - \hat{q}_i$ = conditional probability of remaining HIV-free
- $\hat{S}(t_i) = \prod_{j \le i} \hat{p}_j$ = cumulative HIV-free probability
- $\hat{f}_i = \hat{S}(t_{i-1}) \cdot \hat{q}_i / h_i$ = density estimate
- $\hat{H}_i = 2\hat{q}_i/[h_i(2 - \hat{q}_i)]$ = hazard rate estimate

Table 10: Cohort Life Table for HIV Incidence

Interval	n_i	d_i	w_i	n'_i	$\hat{q}_i$	$\hat{p}_i$	$\hat{S}(t)$	$\hat{f}(t)$	$\hat{H}(t)$	SE($\hat{S}$)
0–2	100	2	3	98.5	0.0203	0.9797	0.9797	0.0102	0.0103	0.0142
2–4	95	1	2	94.0	0.0106	0.9894	0.9693	0.0052	0.0053	0.0175
4–6	92	4	8	88.0	0.0455	0.9545	0.9252	0.0220	0.0233	0.0272
6–8	80	3	10	75.0	0.0400	0.9600	0.8882	0.0185	0.0204	0.0335
8–10	67	2	18	58.0	0.0345	0.9655	0.8576	0.0153	0.0175	0.0387
10–12	47	2	21	36.5	0.0548	0.9452	0.8106	0.0235	0.0282	0.0488
12–14	24	3	21	13.5	0.2222	0.7778	0.6305	0.0901	0.1250	0.0993

Key observations:

- The **median HIV-free time is not reached**: $\hat{S}(14) = 0.6305 > 0.50$.
- Estimated 14-year cumulative incidence: $1 - 0.6305 = 0.3695$ ($\approx$37%).
- Hazard rate is relatively stable at 0.01–0.03/year during years 0–12, then jumps to $\hat{H} = 0.125$/year in the 12–14 interval. However, this last estimate is unreliable ($n' = 13.5$, SE $= 0.0993$).
- Heavy loss to follow-up (83 of 100) substantially reduces precision in later intervals.